Marc Lally

Senior Software Engineer

Platform & Architecture Lead

Core Expertise

Distributed Systems | Serverless | TypeScript/Node

B.Sc. Computer Science (Hons)

UCL - University College London

1990 - 1993

2:1

Distributed Systems

Scalable, high-throughput production environments handling millions of requests

Platform Strategy

Service-oriented and serverless architecture design with operational excellence

Multi-Stack

TypeScript/Node, Python, Rust - selecting the right tool

AWS Solutions Architect Associate

Amazon Web Services

2023

AWS Developer Associate

Amazon Web Services

2023

TLDR; Senior Engineer | Distributed, State-Driven Systems | TypeScript · Node · React · Python · Rust | PostgreSQL · Redis · MongoDB · GraphQL | AI-Embedded Workflows & DevOps | Terraform · Cursor · Warp · Claude · Serverless | Docker | Likes Challenges

As a Senior Software Engineer and Platform & Architecture Lead, I design and operate distributed systems that form the backbone of high-growth companies. My approach centers on translating complex business constraints into elegant, durable technical solutions that scale reliably while remaining operationally sustainable. I've led cross-functional teams through architectural transformations, established platform-wide standards, and consistently elevated engineering maturity across organizations.

Technical Scope & Impact

Design and evolve scalable distributed systems optimizing for reliability, cost efficiency, and velocity. Lead service-oriented and serverless architecture with clear boundaries, data ownership models, and operational visibility. Establish API standards (REST/GraphQL) with rigorous versioning and governance. Code at impact: TypeScript/Node for iteration, Python for data systems, Rust for performance-critical components. Own data architecture across SQL and NoSQL systems, design AWS-native platforms (Lambda, API Gateway, DynamoDB), and establish observability, CI/CD, and security standards at the platform level.

Leadership

Drive architectural direction across teams through data-driven strategy and clear decision frameworks. Mentor senior engineers and emerging technical leads, elevating decision quality and developing the next generation. Bridge engineering, product, and business stakeholders, translating requirements into coherent technical direction and credible delivery plans. Enable teams to move at velocity without sacrificing system integrity - ensuring short-term decisions align with long-term platform health and organizational scalability.

EXPERIENCE

February 2025 - January 2026

CONTRACT

Node.js | TypeScript | Serverless | OAuth 2.0 | JWT

- (NDA-protected work) Engineered low-latency, highly available Node.js/TypeScript microservices on AWS serverless (Lambda, API Gateway, DynamoDB) with OAuth 2.0 and JWT-protected identity flows handling millions of daily requests.
- Established test-driven development culture across engineering squads, reducing production regressions by 60% with comprehensive test automation and 85%+ code coverage.
- Integrated security and compliance into development workflows through secure-by-default patterns and threat modeling, enabling rapid feature velocity without regulatory compromise.
- Mentored junior and mid-level engineers through code reviews, architectural guidance, and workshops on system design and debugging techniques.
- Optimized critical path latency, reducing p99 response times by 40% and improving throughput by 50% through profiling, caching strategies, and infrastructure tuning.
- Implemented comprehensive observability with structured logging, distributed tracing, and custom metrics enabling incident resolution in minutes instead of hours.

March 2023 - December 2024

CONTRACT

React | TypeScript | Node.js | Serverless | OAuth/JWT | TDD

- Architected enterprise-grade backend services in Node.js/TypeScript on serverless platforms handling millions of daily API requests with 99.99% uptime and OAuth/JWT authentication at scale.
- Designed accessible React interfaces meeting WCAG 2.1 AA standards with comprehensive testing across mobile, tablet, and desktop platforms.
- Championed TDD practices and built reusable component libraries with Storybook, accelerating feature velocity while maintaining code quality.
- Led technical design reviews and API governance across the full stack, establishing linting, formatting, and testing standards.
- Mentored engineers on JavaScript patterns, system design, and performance optimization; conducted 50+ technical interviews.
- Implemented monitoring and alerting across services, improving throughput by 50% and reducing latency by 45%.

July 2022 - February 2023

CONTRACT

Node.js | Serverless | AWS Lambda | GraphQL | OAuth/JWT | Payments

- Architected horizontally-scalable Node.js/serverless services with GraphQL schemas supporting complex queries and batching. Implemented OAuth/JWT scopes and role-based access controls handling 50M+ daily transactions.
- Engineered payment integrations with multiple gateways, implemented robust webhook retry logic, idempotency mechanisms, and PCI DSS Level 1 compliance.
- Distilled complex requirements into maintainable architectures with emphasis on stability, observability, and graceful degradation using circuit breakers and bulkheads.
- Implemented comprehensive monitoring with CloudWatch, X-Ray, and custom dashboards; configured intelligent alerting enabling rapid incident resolution and proactive issue detection.
- Mentored 6+ engineers on distributed systems, payment processing, and system design; conducted 40+ technical interviews.
- Improved deployment velocity enabling 5+ releases daily through CI/CD automation and feature flags; conducted disaster recovery drills.

June 2021 - July 2022

Node.js | Nuxt.js | Serverless | AWS | OAuth/JWT | Payments

- Led cross-functional feature delivery by collaborating with product and design teams; shaped information architecture and API contract design that scaled across millions of user sessions.
- Engineered full-stack applications combining Node.js/serverless backends with Nuxt.js frontends on AWS, implementing OAuth/JWT security and complex payment workflows with robust error handling.
- Drove engineering excellence by establishing code review standards, enforcing testing practices, and building internal documentation; grew team capacity through hands-on mentoring and technical pairing.
- Implemented comprehensive monitoring and alerting; analyzed performance metrics to identify bottlenecks and executed targeted optimizations that improved throughput by 50%.
- Resolved critical production incidents through systematic debugging and root cause analysis; established post-incident review practices to prevent recurrence.

December 2020 - June 2021

CONTRACT

GraphQL | Next.js | Node.js | TypeScript | TDD | DV360

- Refactored legacy codebases removing technical debt and security vulnerabilities. Improved test coverage from 40% to 85% across critical payment paths. Built GraphQL/Next.js middleware integrating Google DV360 API handling billions of ad impressions daily.
- Optimized backend and frontend performance through profiling and APM tools. Reduced bundle sizes by 35% and page load times by 45% while maintaining >80% test coverage.
- Implemented security controls protecting sensitive data through end-to-end encryption, field-level access controls, and audit logging. Achieved 100% secure code review compliance.
- Architected modular systems with clear separation of concerns. Created reusable utilities that accelerated engineer onboarding from 2 weeks to 4 days.
- Collaborated with frontend and data engineering teams on GraphQL schema design and API contracts. Conducted architecture walkthroughs and technical training.
- Optimized PostgreSQL queries and schema design; implemented connection pooling and strategic indexing improving performance by 50%.

December 2019- November 2020

Node.js | Serverless | OAuth/JWT | Payments | POS | SQL/NoSQL

- Orchestrated complex integrations connecting ordering, POS systems (NCR, MICROS), payment processors, and loyalty platforms. Established API boundaries with contract testing preventing integration failures.
- Designed Node.js/serverless APIs on AWS Lambda with OAuth/JWT authentication and multi-tenancy. Implemented PCI DSS Level 1 compliance with 99.99% uptime.
- Optimized PostgreSQL and DynamoDB for <50ms p95 latency. Implemented multi-layer caching improving performance by 70%.
- Coordinated delivery across distributed teams spanning 5 time zones. Enabled 50+ monthly feature releases with zero production incidents.
- Mentored 6 engineers on distributed systems and payment processing; conducted 35+ technical interviews.
- Optimized Lambda cold starts from 3s to 500ms and reduced AWS costs by 40% while improving performance by 30%.

February 2019 - November 2019

CONTRACT

React | Vanilla JS | ElasticSearch | TypeScript | TDD

- Owned core front-end platform serving 5,000+ institutional investors globally; shipped high-impact React and Vanilla JS features integrated with Elasticsearch backend enabling complex financial data search and visualization.
- Authored technical specifications and architectural decision records for applications; advocated for user-centered design, accessibility (WCAG 2.1 AA), and performance-first implementation.
- Pioneered performance optimization, reducing bundle size by 35% and improving Core Web Vitals (LCP 2.1s→1.2s, CLS 0.15→0.05). Established performance monitoring with real user metrics.
- Implemented comprehensive testing with Jest, React Testing Library, and visual regression testing. Achieved 85%+ code coverage across critical financial data flows.
- Mentored 4 junior engineers on modern JavaScript, React hooks, TypeScript, accessibility, and system design principles through code reviews and technical workshops.
- Engineered Elasticsearch integration enabling full-text search across financial taxonomies; reduced search latency from 800ms to 150ms p95.

January 2018 - December 2018

CONTRACT

React | Node.js | Serverless | AWS | CMS

- Engineered end-to-end CMS platform using React and Node.js on AWS serverless; enabled content teams to publish VR experiences across multiple distribution channels from unified interface.
- Designed component-driven front-end architecture with Storybook (100+ components) and design tokens ensuring consistency and accelerating feature delivery by 40%.
- Implemented resilient data integration pipelines handling 10,000+ daily content updates; managed schema evolution and zero-downtime deployments.
- Established CI/CD automation with GitHub Actions and infrastructure-as-code; reduced deployment time from 4 hours to 15 minutes enabling 16x faster shipping.
- Optimized performance reducing page load time from 3.5s to 1.2s through code splitting, lazy loading, and caching; implemented monitoring with New Relic.
- Established testing practices with Jest, Supertest, and load testing; achieved 80%+ coverage catching 95% of bugs in CI.

April 2017 - November 2017

CONTRACT

Node.js | Koa | Swagger | PostgreSQL | MongoDB | Docker | Stripe

- Architected microservices platform for mental health counseling serving 50,000+ users; designed clear service boundaries with OpenAPI/Swagger contracts ensuring loose coupling and independent deployment.
- Managed polyglot persistence (PostgreSQL, MongoDB, Redis); optimized queries and designed indexes achieving 99.5% uptime with <50ms p95 latency.
- Integrated Stripe payment processing with PCI DSS compliance; processed $10M+ annually with zero fraud incidents and 99.99% success rate.
- Pioneered test automation with Jest, Supertest, and Pact; achieved 80%+ coverage on critical payment flows preventing regressions.
- Containerized platform with Docker and Kubernetes; automated infrastructure with Terraform reducing deployment time from 4 hours to 15 minutes.
- Optimized performance reducing p99 latency from 800ms to 120ms and infrastructure costs by 35% through profiling and caching.

October 2016 - March 2017

CONTRACT

PHP | OAuth2 | SAML | Craft CMS | MySQL | jQuery

- Spearheaded OAuth 2.0 and SAML 2.0 integrations enabling SSO across 50+ enterprise client ecosystems with robust key rotation and token refresh mechanisms.
- Developed custom Craft CMS plugins (20+ extensions) enabling client-specific workflows with PHP, MySQL, and jQuery.
- Introduced comprehensive test coverage across legacy codebase; reduced production incidents by 65%.
- Modernized legacy systems through incremental refactoring; maintained backwards compatibility and zero-downtime migrations.
- Mentored 3 developers on authentication patterns and enterprise integration through pair programming.
- Optimized MySQL for CMS platform serving 100+ sites; reduced page load time from 2.5s to 800ms.

January 2015 - September 2016

CONTRACT

PHP | MySQL | LAMP | PHPUnit | Docker | CI/CD

- Led delivery of complex enterprise web applications on LAMP stack; established engineering standards including code style guides and security policies reducing incidents by 80%.
- Designed normalized data models and RESTful APIs handling complex educational content management and regulatory compliance for UK schools.
- Established PHPUnit testing and TDD; drove coverage from 10% to 70% and reduced production bugs by 50%.
- Pioneered Docker and Jenkins CI/CD; reduced deployment time from 2 hours to 10 minutes enabling daily releases.
- Mentored engineering team of 6 through code reviews, interviews, and design sessions focused on architectural thinking.
- Optimized MySQL for 100,000+ concurrent users; improved load times from 3.2s to 420ms p99 through indexing and caching.

May 2006 - October 2014

PHP | Laravel | MySQL | jQuery | Architecture

- Built and maintained custom enterprise software and system integrations for financial services, healthcare, and e-commerce clients delivering solutions across diverse domains.
- Owned end-to-end requirements discovery and specification; produced technical architecture diagrams and design documents aligned with business objectives.
- Systematically optimized legacy system performance; implemented database indexing, query optimization, and caching improving response times by 60%.
- Provided 24/7 production support with critical issue resolution; maintained 99.9% uptime SLAs and built operational runbooks.
- Mentored multiple generations of developers on Laravel, database design, API development, and debugging techniques; conducted 100+ interviews.
- Designed and optimized complex schemas for diverse domains; implemented proper normalization and backup procedures ensuring data integrity.

March 2001 - February 2006

Reporting | Automation | Data Strategy | Operations

- Engineered technical integrations and automated reporting for telecommunications services serving 10 European countries with 50M+ customers; connected disparate legacy systems enabling real-time data visibility.
- Established development standards and code review practices across operations teams, reducing defects by 35%.
- Spearheaded digital transformation from manual processes to automated workflows; built tools and dashboards reducing administrative overhead by 40%.
- Designed data warehouses and reporting schemas enabling business intelligence for C-suite strategic decisions.
- Mentored junior developers on troubleshooting and database optimization; conducted technical interviews.
- Maintained 24/7 on-call availability for mission-critical billing systems resolving critical issues within SLAs.

1997 - 2001

Online Banking | Web Apps | Security | Compliance

- Developed mission-critical online banking software with stringent security requirements; implemented SSL encryption, input validation, and threat modeling.
- Ensured regulatory compliance with banking standards (BSA, AML, KYC); embedded controls into product design and testing.
- Facilitated cross-functional collaboration across product, systems, QA, operations, and risk teams; shipped features maintaining 99.99% uptime.
- Supported full lifecycle from requirements through production; maintained 24/7 on-call rotation for critical incident resolution.
- Developed core banking features (transaction processing, account management, reporting) with rigorous QA and security testing.
- Optimized complex database systems storing terabytes of transactional data; ensured data integrity and backup/recovery.

1994 - 1997

Web Apps | Security | Reliability | SDLC

- Built customer account and transaction web applications for Next's online division; implemented transactional integrity and comprehensive audit logging.
- Contributed across full lifecycle including requirements analysis, design, implementation, testing, and deployment meeting regulatory timelines.
- Implemented security best practices including password hashing, secure session management, and encryption ensuring PCI compliance.
- Supported production systems through rapid incident response; participated in 24/7 on-call rotation for mission-critical banking infrastructure.
- Learned fundamental software engineering principles in disciplined corporate environment; developed expertise in requirements-driven development and compliance-first design.
- Designed database schemas for transaction systems; understood ACID properties and data consistency requirements for critical systems.

CONNECT